26 Yes/No Questions Explained

Entropy, spontaneity, heat engines,
and thermodynamic potentials

逐题讲解与教材式公式推导

This guide preserves all 26 questions, in their original order, from the currently displayed Notebook quiz *"3 Quiz yes/no entropy"* in *"Thermodynamics, Ch. 1,2,3."*

Each question has its Yes/No answer, a derivation, the assumptions needed to apply the equations, and a short exam reminder. Important technical terms have Chinese annotations. Numerical examples are illustrative unless identified otherwise.

The answers follow the quiz's intended introductory-thermodynamics setting. Where its wording or feedback omits a condition, the explanation states it explicitly. In particular, read Questions 3, 7, 13, 15, 16, 21, and 26 carefully.

Prepared for Jiasheng Fu 28 October 2026

Source: the visible quiz and its revealed answer feedback; explanations independently developed here. The seven source PDFs listed by the notebook were not separately read. Supplementary references appear at the end.

Answer key and working conventions

Question	Answer	Question	Answer	Question	Answer	Question	Answer
1	Yes	8	Yes	15	No	22	No
2	No	9	No	16	Yes	23	Yes
3	No	10	Yes	17	Yes	24	Yes
4	Yes	11	No	18	No	25	Yes
5	No	12	Yes	19	Yes	26	No
6	Yes	13	Yes	20	Yes		
7	Yes	14	Yes	21	No		

Heat and work signs. Heat entering the system is positive. For system work, the chemistry convention is $\Delta U = q + w_{\text{on}}$. In engine calculations, $W_{\text{out}} > 0$ is work delivered; Q_h, Q_c are positive heat magnitudes. All temperatures in ratios and entropy denominators are kelvin.

Notation. Subscripts sys, surr, and tot mean system, surroundings, and their isolated total. S_{gen} denotes entropy production. C is a total heat capacity; C_{m} is molar. A superscript $\circ$ denotes the standard state, equivalent here to the quiz's $\ominus$ notation. ln is the natural logarithm.

Three equations to keep distinct.

$$\Delta S_{\text{sys}} = \int_i^f \frac{\delta q_{\text{rev}}}{T}, \qquad \text{equilibrium endpoints; reversible reference path,}$$

$$\Delta S_{\text{surr}} = -\frac{q_{\text{sys}}}{T_{\text{surr}}}, \qquad \text{ideal constant-temperature reservoir,}$$

$$\Delta S_{\text{tot}} \geq 0, \qquad \text{isolated combined total.}$$

The first equation defines an entropy difference; the second describes the ideal reservoir; the third imposes the direction of an allowed macroscopic change. Actual irreversible heat cannot generally replace δq_{rev} in the first equation.

Reading route. Questions 1–5 establish the entropy rules; 6–7 give the statistical picture; 8–17 develop thermodynamic tests and calorimetry; 18–26 consolidate applications and common traps. Every question begins on its own page, and PDF bookmarks provide direct navigation.

QUESTION 1 / 26

The second law and an isolated system

Is the total change in entropy of an isolated system always non-negative for any spontaneous physical or chemical process?

Answer: Yes

An isolated system （孤立系统） exchanges neither matter nor energy with its surroundings. There is therefore no entropy transfer across its boundary; its entropy change equals its entropy production （熵产生）.

$$\Delta S_{\text{isolated}} = S_{\text{gen}},$$
$$S_{\text{gen}} \geq 0.$$

For a spontaneous irreversible change, the inequality is strict. Equality describes the reversible limit or a system that remains at equilibrium （平衡）.

$$\boxed{\Delta S_{\text{isolated}} > 0 \quad \text{for an irreversible spontaneous change.}}$$

For example, heat flowing internally from a hot body to a colder body produces entropy even though no energy enters the combined isolated pair.

$$\Delta S_{\text{tot}} = Q\left(\frac{1}{T_c} - \frac{1}{T_h}\right) > 0, \qquad T_h > T_c, \quad Q > 0.$$

Remember. The second law constrains the isolated total, not necessarily the entropy of an individual subsystem. “Spontaneous” describes thermodynamic direction; it does not imply a fast reaction.

QUESTION 2 / 26

Why a Carnot engine cannot reach 100 percent

Can a Carnot heat engine operating between a hot source at T_{hot} and a cold sink at $T_{cold} > 0$ K achieve an efficiency of 100%?

Answer: No

Let $Q_h > 0$ be the heat absorbed from the hot reservoir, $Q_c > 0$ the heat rejected to the cold reservoir, and $W_{\text{out}} > 0$ the work delivered per cycle. For a reversible Carnot cycle,

$$\frac{Q_h}{T_h} = \frac{Q_c}{T_c},$$
$$W_{\text{out}} = Q_h - Q_c,$$
$$\eta_{\text{C}} = \frac{W_{\text{out}}}{Q_h} = 1 - \frac{Q_c}{Q_h} = 1 - \frac{T_c}{T_h}.$$

With finite $T_h > T_c > 0$, the ratio T_c/T_h is positive. Thus

$$\boxed{\eta \leq \eta_{\text{C}} < 1.}$$

For $T_h = 600\,\text{K}$ and $T_c = 300\,\text{K}$, even a reversible engine has $\eta_{\text{C}} = 0.50$. Irreversibility （不可逆性）can only reduce the efficiency below this limit.

Remember. Use absolute temperatures in kelvin. A temperature of $0\,^\circ\text{C}$ is $273.15\,\text{K}$, not an absolute-zero sink.

QUESTION 3 / 26

The surroundings as a thermal reservoir

Does the calculation of the entropy change of the surroundings, ΔS_{surr}, depend on whether the process taking place in the system is reversible or irreversible?

Answer: No

This is the quiz's intended answer under the thermal-reservoir assumption （恒温热库假设）. The surroundings are taken to remain at a uniform fixed temperature T_{surr} and to have no additional internal entropy production.

$$\begin{aligned} \Delta S_{\text{surr}} &= \frac{q_{\text{surr}}}{T_{\text{surr}}}, \\ q_{\text{surr}} &= -q_{\text{sys}}, \\ \therefore \quad \Delta S_{\text{surr}} &= -\frac{q_{\text{sys}}}{T_{\text{surr}}}. \end{aligned}$$

The same reservoir formula applies whether the system changes reversibly or irreversibly. Heat crossing a finite temperature difference can produce entropy in the combined system and surroundings.

The statement does *not* mean that every possible path transfers the same heat. Heat is a path function （路径函数）; changing the path can change q_{sys} and consequently the numerical value of ΔS_{surr}.

Remember. "Independent of reversibility" refers to the reservoir calculation rule. For finite, non-isothermal surroundings, use a suitable entropy integral or entropy balance instead of one fixed q/T.

QUESTION 4 / 26

Isothermal expansion of an ideal gas

Is the change in entropy of a perfect gas undergoing an isothermal expansion from volume V_i to V_f given by $\Delta S = nR\ln(V_f/V_i)$?

Answer: Yes

For a fixed amount n of an ideal gas （理想气体）, internal energy depends only on temperature. An isothermal process therefore has $\Delta U = 0$. Choose a reversible reference path between the same equilibrium endpoints.

$$\begin{aligned} p &= \frac{nRT}{V}, \\ \delta q_{\mathrm{rev}} &= p\,\mathrm{d}V, \\ \Delta S &= \int_{V_i}^{V_f} \frac{\delta q_{\mathrm{rev}}}{T} = nR\int_{V_i}^{V_f} \frac{\mathrm{d}V}{V} \\ &= \boxed{nR\ln\left(\frac{V_f}{V_i}\right)}. \end{aligned}$$

Entropy is a state function （状态函数）, so the result also describes an irreversible isothermal expansion between these endpoints. In insulated free expansion of an ideal gas, actual $q = 0$ but $\Delta S > 0$; substituting actual heat into q/T would fail.

For one mole expanding to twice its original volume,

$$\Delta S = R\ln 2 \approx 5.76\,\mathrm{J\,K^{-1}}.$$

Remember. The subscript “rev” belongs to the heat used to compute entropy, not necessarily to the actual process.

QUESTION 5 / 26

Equal heat has a larger entropy effect at lower temperature

Is the transfer of a specific amount of energy as heat into a system at low temperature accompanied by a smaller entropy change than transferring the same heat at high temperature?

Answer: No

For reversible heat transfer （可逆传热） at a fixed temperature,

$$\Delta S = \frac{q_{\mathrm{rev}}}{T}.$$

If the same positive heat q is added at two temperatures $T_1 < T_2$, then

$$\frac{q}{T_1} > \frac{q}{T_2}.$$

For an illustrative 100 J transfer,

$$\Delta S_{250\,\mathrm{K}} = \frac{100}{250} = 0.400\,\mathrm{J\,K^{-1}},$$
$$\Delta S_{500\,\mathrm{K}} = \frac{100}{500} = 0.200\,\mathrm{J\,K^{-1}}.$$

The colder receiving body undergoes the larger entropy increase for the same reversible heat input. The thermodynamic weighting of heat is $1/T$.

If the body's temperature changes appreciably while it is heated, a single temperature cannot describe the entire process:

$$\Delta S = \int_{T_i}^{T_f} \frac{C(T)}{T}\,\mathrm{d}T.$$

Remember. The comparison assumes positive heat input and a reversible reference path or an equilibrium reservoir. For arbitrary irreversible heating, actual heat divided by system temperature is not a general entropy formula.

QUESTION 6 / 26

A single microstate and the Boltzmann formula

According to the Boltzmann formula $S = k \ln W$, is the statistical entropy equal to zero when a system has only a single accessible microstate ($W = 1$)?

Answer: Yes

For W equally probable accessible microstates （微观态）, Boltzmann's expression is

$$S = k_{\mathrm{B}} \ln W.$$

If exactly one microstate is available,

$$W = 1 \quad \Longrightarrow \quad S = k_{\mathrm{B}} \ln 1 = \boxed{0}.$$

The more general Gibbs statistical entropy （吉布斯统计熵） uses probabilities p_i:

$$S = -k_{\mathrm{B}} \sum_i p_i \ln p_i.$$

With one certain state, $p_1 = 1$, its contribution is zero. For equally probable states, $p_i = 1/W$, the Gibbs expression reduces to $k_{\mathrm{B}} \ln W$.

This is a statement about the statistical description and its accessible-state count. Merely knowing the exact microscopic state of a hot sample does not make its thermodynamic entropy zero: the relevant thermodynamic ensemble still contains many possible microstates.

Remember. "Only one accessible microstate" is a physical/statistical constraint. It is stronger than "we have identified one particular microstate."

QUESTION 7 / 26

Residual entropy in proton-disordered ice

Does ice possess a non-zero residual entropy at $T = 0$ K due to structural randomness in its hydrogen bonding network?

Answer: Yes

The intended example is proton-disordered ice （质子无序冰）, whose oxygen lattice can be crystalline while many hydrogen arrangements satisfy the ice rules. The number of configurations remains larger than one when this disorder is frozen in.

$$S_{\mathrm{res}} = k_{\mathrm{B}} \ln W_0 > 0.$$

Pauling's approximate count for N water molecules is

$$\begin{aligned} W_0 &\approx \left(\frac{3}{2}\right)^N, \\ S_{\mathrm{res,m}} &\approx R \ln\left(\frac{3}{2}\right) \\ &\approx 3.37\,\mathrm{J\,K^{-1}\,mol^{-1}}. \end{aligned}$$

This explains the quiz's quoted value of roughly $3.4\,\mathrm{J\,K^{-1}\,mol^{-1}}$. It is configurational entropy （构型熵）, rather than a claim that ordinary thermal vibrations persist at zero temperature.

Qualification. The answer concerns the disordered ice sample and its frozen configurations. It is not a claim that every equilibrium ice phase must have residual entropy; proton-ordered ice XI is the contrasting low-temperature phase. The third-law zero applies to an appropriate perfectly ordered crystal with a non-degenerate ground state. [R1]

Remember. A crystalline oxygen framework does not by itself imply complete proton ordering.

QUESTION 8 / 26

Helmholtz energy at constant temperature and volume

Is a decrease in Helmholtz energy ($A = U - TS$) the fundamental criterion for spontaneity for a process occurring at constant temperature and volume?

Answer: Yes

Helmholtz energy （亥姆霍兹自由能） is

$$A = U - TS.$$

Consider a closed system in a thermal reservoir, with fixed T, V and only pressure-volume work. Because the volume is fixed, there is no boundary expansion work, so $q_{\text{sys}} = \Delta U$.

$$\begin{aligned} \Delta S_{\text{tot}} &= \Delta S_{\text{sys}} - \frac{\Delta U}{T}, \\ -T\Delta S_{\text{tot}} &= \Delta U - T\Delta S_{\text{sys}} \\ &= \Delta A. \end{aligned}$$

Consequently,

$$\boxed{\Delta S_{\text{tot}} > 0 \iff \Delta A < 0.}$$

At stable equilibrium, A is minimized over the changes allowed by the constraints. For a reversible limiting change, $\Delta S_{\text{tot}} = 0$.

If external electrical or shaft work is supplied, the simple "A must decrease" rule needs the corresponding work term. Imposed work can drive a change that increases the system's free energy.

Remember. Pair Helmholtz energy with constant T, V. State the permitted work interactions before applying a free-energy spontaneity test.

QUESTION 9 / 26

Comparing gaseous and liquid molar entropies

Is the standard molar entropy of a gas typically lower than that of its corresponding liquid phase at the same temperature?

Answer: No

For ordinary vaporization （汽化）, the gas has a higher molar entropy than the liquid. At the equilibrium boiling temperature T_b and the specified pressure,

$$\Delta_{\mathrm{vap}} S_{\mathrm{m}} = S_{\mathrm{m},g} - S_{\mathrm{m},l} = \frac{\Delta_{\mathrm{vap}} H_{\mathrm{m}}}{T_b} > 0.$$

Vaporization is endothermic: $\Delta_{\mathrm{vap}} H_{\mathrm{m}} > 0$. The associated entropy increase reflects the much larger spatial volume and increased number of accessible molecular arrangements in a gas.

For standard molar entropies （标准摩尔熵） at the same temperature, the usual ordering is

$$S^{\circ}_{\mathrm{m},g} > S^{\circ}_{\mathrm{m},l}.$$

The word “typically” matters: compare the same substance under clearly specified standard-state conditions. The equilibrium formula $\Delta S = \Delta H / T_b$ must be used at the phase-transition temperature, not at an arbitrary temperature by substituting an unrelated enthalpy value.

Remember. A gas usually has higher entropy than its corresponding liquid. Entropy differences depend on both phase and thermodynamic state, including pressure.

QUESTION 10 / 26

Entropy balance in an equilibrium phase transition

Is the entropy change of an endothermic phase transition at its equilibrium transition temperature matched by an equal and opposite entropy change in the surroundings?

Answer: Yes

At the equilibrium transition temperature T_{trs} and fixed pressure, heat can be transferred reversibly between a reservoir and the phase-changing sample. With only pressure-volume work,

$$q_{\text{sys}} = \Delta H_{\text{trs}} > 0$$

for an endothermic transition （吸热相变）. The two entropy changes are

$$\begin{aligned} \Delta S_{\text{sys}} &= \frac{\Delta H_{\text{trs}}}{T_{\text{trs}}}, \\ \Delta S_{\text{surr}} &= -\frac{\Delta H_{\text{trs}}}{T_{\text{trs}}}, \\ \therefore \quad \Delta S_{\text{tot}} &= 0. \end{aligned}$$

For example, melting at the melting point increases the sample's entropy while removing exactly the same entropy from a reservoir at that temperature.

The equality is an equilibrium, reversible-limit statement. If the heat source has a higher temperature $T_h > T_{\text{trs}}$, the entropy loss of that source is smaller in magnitude:

$$\Delta S_{\text{tot}} = \Delta H_{\text{trs}} \left(\frac{1}{T_{\text{trs}}} - \frac{1}{T_h} \right) > 0.$$

Remember. Equal and opposite entropy changes require reversible heat exchange at the transition temperature. Merely observing a phase change does not guarantee zero total entropy production.

QUESTION 11 / 26

The Kelvin-Planck restriction on a cyclic engine

Can a cyclic heat engine continuously convert heat completely into mechanical work without discharging any waste heat to a colder reservoir?

Answer: No

The Kelvin-Planck statement （开尔文-普朗克表述） rules out a cyclic engine whose sole effect is to absorb heat from one reservoir and convert all of it into work.

Because entropy is a state function, the working substance returns to its original entropy after a cycle:

$$\Delta S_{\text{engine}} = 0.$$

If it took $Q_h > 0$ from a single reservoir and rejected no heat, the reservoir would lose entropy while work would carry no entropy to compensate:

$$\Delta S_{\text{tot}} = -\frac{Q_h}{T_h} < 0.$$

This contradicts the second law. A conventional two-reservoir engine instead satisfies

$$-\frac{Q_h}{T_h} + \frac{Q_c}{T_c} \geq 0.$$

The wording “cyclic” is essential. In a reversible isothermal expansion of an ideal gas, absorbed heat can equal delivered work during that single step, but the gas has not returned to its initial state. Closing the cycle restores the restriction.

Remember. Complete heat-to-work conversion in one step is possible; complete conversion as the sole net effect of a cycle is not.

QUESTION 12 / 26

The Debye law and vanishing heat capacity

Does the Debye T^3-law predict that the molar constant-pressure heat capacity $C_{p,m}$ of a non-metallic solid approaches zero as temperature approaches absolute zero?

Answer: Yes

In the low-temperature Debye regime （德拜低温区）, the lattice heat capacity of a three-dimensional insulating crystal obeys

$$C_{V,\mathrm{m}} \simeq \frac{12\pi^4 R}{5}\left(\frac{T}{\Theta_D}\right)^3, \qquad T \ll \Theta_D.$$

Here Θ_D is the Debye temperature. At sufficiently low temperature, the difference between C_p and C_V is usually negligible for this purpose, so the quiz uses

$$C_{p,\mathrm{m}} \simeq aT^3,$$
$$\lim_{T\to 0} C_{p,\mathrm{m}} = 0.$$

The factor C_p/T also has a regular low-temperature limit:

$$\frac{C_{p,\mathrm{m}}}{T} \simeq aT^2 \longrightarrow 0.$$

Thus the entropy integral converges at its lower endpoint. Microscopically, fewer phonon modes （声子模） are thermally excited as temperature decreases. [R2]

Remember. The basic Debye result is for C_V and a low-temperature lattice contribution. It is not a universal $C_p = aT^3$ law for all solids or all temperatures; metals can also have an electronic term proportional to T.

QUESTION 13 / 26

Surroundings entropy from the system enthalpy

Is the entropy change of the surroundings for a process occurring at constant pressure given by $\Delta S_{surr} = -\Delta H/T$*?*

Answer: Yes

The intended formula assumes a constant-temperature reservoir, a closed system at constant pressure, and pressure-volume work only. Under these conditions,

$$\begin{aligned} q_{\mathrm{sys},p} &= \Delta H_{\mathrm{sys}}, \\ q_{\mathrm{surr}} &= -q_{\mathrm{sys},p}, \\ \Delta S_{\mathrm{surr}} &= \boxed{-\frac{\Delta H_{\mathrm{sys}}}{T_{\mathrm{surr}}}}. \end{aligned}$$

For an exothermic process （放热过程）, $\Delta H_{\mathrm{sys}} < 0$, and the reservoir gains entropy. For an endothermic process, the reservoir loses entropy.

As an illustrative calculation, releasing $30.0\,\mathrm{kJ}$ to a reservoir at $300\,\mathrm{K}$ gives

$$\Delta S_{\mathrm{surr}} = -\frac{-30\,000\,\mathrm{J}}{300\,\mathrm{K}} = 100\,\mathrm{J\,K^{-1}}.$$

Constant pressure by itself is insufficient to justify the entire expression. Additional electrical work can make $q_p \neq \Delta H$, and finite surroundings may change temperature.

Remember. The denominator is the surroundings' absolute temperature. Do not replace it by a changing system temperature in an irreversible process.

QUESTION 14 / 26

A spontaneous process can reduce system entropy

Can a chemical reaction or physical process occur spontaneously if the system's entropy decreases?

Answer: Yes

The spontaneity test concerns the combined isolated total:

$$\Delta S_{\text{tot}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} > 0.$$

Consequently, a negative system entropy change is allowed if the surroundings gain more entropy than the system loses:

$$\Delta S_{\text{sys}} < 0, \qquad \Delta S_{\text{surr}} > |\Delta S_{\text{sys}}|.$$

Freezing （凝固） below the melting point illustrates the mechanism. If the latent heat released is approximately $L > 0$ and the surroundings are at $T < T_m$, then approximately

$$\begin{aligned} \Delta S_{\text{sys}} &\simeq -\frac{L}{T_m}, \\ \Delta S_{\text{surr}} &\simeq \frac{L}{T}, \\ \Delta S_{\text{tot}} &\simeq L\left(\frac{1}{T} - \frac{1}{T_m}\right) > 0. \end{aligned}$$

This simple illustration neglects heat-capacity corrections; the sign argument is the point. Ordering within a sample can be accompanied by a larger entropy increase outside it.

Remember. Do not infer "nonspontaneous" from $\Delta S_{\text{sys}} < 0$ alone. Identify the surroundings and include their entropy change.

QUESTION 15 / 26

Work carries no entropy, but dissipation can generate it

Is energy transferred as work considered a form of energy flow that directly changes entropy by using disorderly molecular motion?

Answer: No

Work （功） is energy transfer through a generalized force and displacement, such as a piston moving or a shaft turning. It does not directly carry entropy across a system boundary in the way heat does.

For a closed system, the entropy balance is

$$\Delta S = \int \frac{\delta q}{T_b} + S_{\text{gen}}, \qquad S_{\text{gen}} \geq 0,$$

where T_b is the boundary temperature at the heat-transfer location. There is no separate "entropy carried by work" term.

Important correction to the short quiz explanation. It does not follow that doing work can never change entropy. Stirring an insulated liquid dissipates mechanical energy into internal energy. For an adiabatic （绝热） process,

$$q = 0,$$
$$\Delta S = S_{\text{gen}} > 0$$

when friction or viscous dissipation is present. A reversible adiabatic process, by contrast, has $S_{\text{gen}} = 0$ and is isentropic （等熵）.

Remember. Distinguish entropy transfer from entropy production. Work transfers energy without transferring entropy; irreversible conversion of work into thermal energy can still produce entropy.

QUESTION 16 / 26

Third-law entropy as an area, with phase changes included

Is the third-law standard molar entropy $S^{\ominus}(T)$ calculated by finding the area under the plot of C_p/T versus T from 0 K to temperature T?

Answer: Yes

Within a single phase at constant pressure and fixed composition,

$$\mathrm{d}S^{\circ}_{\mathrm{m}} = \frac{C^{\circ}_{p,\mathrm{m}}(T)}{T}\,\mathrm{d}T.$$

If no phase transition intervenes and the perfectly ordered reference crystal has $S_{\mathrm{m}}(0) = 0$, then

$$S^{\circ}_{\mathrm{m}}(T) = \int_0^T \frac{C^{\circ}_{p,\mathrm{m}}(T')}{T'}\,\mathrm{d}T'.$$

This is the area under a plot of $C^{\circ}_{p,\mathrm{m}}/T$ against T. The dummy variable T' keeps the upper limit distinct from the integration variable.

For a real heating path crossing phase transitions, the complete expression is

$$\boxed{S^{\circ}_{\mathrm{m}}(T) = S_{\mathrm{res,m}} + \sum_{\text{phases}} \int \frac{C^{\circ}_{p,\mathrm{m}}(T')}{T'}\,\mathrm{d}T' + \sum_{j} \frac{\Delta_{\mathrm{trs}} H^{\circ}_{\mathrm{m},j}}{T_{\mathrm{trs},j}}.}$$

Each phase integral uses its own temperature interval. Latent heat （潜热） contributes an entropy jump; an ordinary smooth heat-capacity curve alone does not include that jump.

Remember. Answer Yes to the intended area interpretation, but include transition entropies and any residual entropy when calculating an absolute value.

QUESTION 17 / 26

Gibbs energy and spontaneity at constant temperature and pressure

At constant temperature and pressure, is a chemical reaction spontaneous if the change in Gibbs energy, ΔG, is negative?

Answer: Yes

Gibbs energy （吉布斯自由能） is

$$G = H - TS.$$

For a closed system at constant T, p, with only pressure-volume work and a reservoir at the same fixed temperature,

$$\begin{aligned} \Delta S_{\text{tot}} &= \Delta S_{\text{sys}} - \frac{\Delta H}{T}, \\ \Delta G &= \Delta H - T\Delta S_{\text{sys}} \\ &= -T\Delta S_{\text{tot}}. \end{aligned}$$

Since $T > 0$,

$$\boxed{\Delta G < 0 \iff \Delta S_{\text{tot}} > 0.}$$

For a reaction at its current composition, the local forward-direction test uses reaction Gibbs energy （反应吉布斯能）:

$$\mathrm{d}G = \Delta_r G \,\mathrm{d}\xi, \qquad \Delta_r G = \Delta_r G^\circ + RT \ln Q.$$

Here ξ is reaction extent （反应进度） and Q is the dimensionless reaction quotient. A negative standard value alone does not determine the direction at every composition.

Remember. Use the actual $\Delta_r G$ for reaction direction. A negative value predicts a thermodynamic driving force, not a reaction rate or the absence of an activation barrier.

QUESTION 18 / 26

Isothermal compression decreases ideal-gas entropy

When a sample of gas is compressed isothermally to a smaller volume ($V_f < V_i$), does the entropy of the gas sample increase?

Answer: No

Use the ideal-gas entropy relation for fixed n:

$$\Delta S = nR \ln\left(\frac{V_f}{V_i}\right).$$

For compression, $0 < V_f < V_i$, so

$$0 < \frac{V_f}{V_i} < 1,$$
$$\ln\left(\frac{V_f}{V_i}\right) < 0,$$
$$\therefore \quad \Delta S < 0.$$

For one mole compressed to half its original volume,

$$\Delta S = R \ln(1/2) = -R \ln 2 \approx -5.76\,\mathrm{J\,K^{-1}}.$$

In a reversible isothermal compression, heat leaves the gas, and the surroundings gain exactly the entropy lost by the gas. In an irreversible compression, total entropy production is positive even though the gas's entropy change between the same endpoints is still negative.

The quiz uses the ideal-gas model. For a general real fluid, volume dependence follows a thermodynamic derivative and should not automatically be replaced by nR/V.

Remember. The negative entropy change belongs to the gas. It does not imply a negative total entropy change for the gas plus surroundings.

QUESTION 19 / 26

Connecting microscopic and molar constants

Does the Boltzmann constant k equal the gas constant R divided by Avogadro's constant N_A?

Answer: Yes

The Boltzmann constant （玻尔兹曼常数） is the particle-scale counterpart of the molar gas constant:

$$R = N_{\mathrm{A}} k_{\mathrm{B}}, \qquad \boxed{k_{\mathrm{B}} = \frac{R}{N_{\mathrm{A}}}}.$$

Their SI values are

$$\begin{aligned} k_{\mathrm{B}} &= 1.380649 \times 10^{-23}\,\mathrm{J\,K^{-1}}, \\ N_{\mathrm{A}} &= 6.02214076 \times 10^{23}\,\mathrm{mol^{-1}}, \\ R &\approx 8.314462618\,\mathrm{J\,mol^{-1}\,K^{-1}}. \end{aligned}$$

The first two constants have exact defining values in the SI. Dividing a per-mole constant by the number of entities per mole produces a particle-scale constant:

$$\frac{\mathrm{J\,mol^{-1}\,K^{-1}}}{\mathrm{mol^{-1}}} = \mathrm{J\,K^{-1}}.$$

This is why the ideal-gas equation has two equivalent forms,

$$pV = N k_{\mathrm{B}} T = nRT, \qquad N = n N_{\mathrm{A}}.$$

Remember. Use k_{B} with a number of particles N, and R with an amount of substance n in moles. Do not count the factor N_{A} twice.

QUESTION 20 / 26

Entropy gained by heating at fixed volume

Is the entropy change for heating a substance at constant volume calculated using the constant-volume heat capacity C_V in $\Delta S = \int \frac{C_V}{T} dT$?

Answer: Yes

Take a closed, fixed-composition sample at constant volume, within one phase. With no non-expansion work, choose a reversible heating path:

$$\begin{aligned} \delta q_{\mathrm{rev},V} &= \mathrm{d}U = C_V(T)\,\mathrm{d}T, \\ \mathrm{d}S &= \frac{\delta q_{\mathrm{rev},V}}{T}, \\ \Delta S &= \boxed{\int_{T_i}^{T_f} \frac{C_V(T)}{T}\,\mathrm{d}T}. \end{aligned}$$

If the total heat capacity is approximately temperature independent,

$$\Delta S = C_V \ln\!\left(\frac{T_f}{T_i}\right).$$

For molar heat capacity （摩尔热容）, include the amount of substance:

$$\Delta S = n \int_{T_i}^{T_f} \frac{C_{V,\mathrm{m}}(T)}{T}\,\mathrm{d}T.$$

The formula computes the entropy difference even if the actual heating was irreversible, provided the same equilibrium endpoints are used. If a phase change occurs, its contribution must be handled consistently rather than assuming one smooth heat-capacity law throughout.

Remember. Choose C_V for fixed-volume heating and C_p for fixed-pressure heating. Distinguish total heat capacity from molar heat capacity before inserting n.

QUESTION 21 / 26

A refrigerator needs work and obeys the second law

Does the operation of a refrigerator violate the Second Law of Thermodynamics because heat flows from a cold region to a warm region?

Answer: No

Let a cyclic refrigerator remove $Q_c > 0$ from a cold reservoir and reject $Q_h > 0$ to a hot reservoir. Its work input is

$$W_{\text{in}} = Q_h - Q_c > 0.$$

The reservoir entropy changes satisfy

$$\Delta S_{\text{tot}} = \frac{Q_h}{T_h} - \frac{Q_c}{T_c} \geq 0.$$

For a reversible refrigerator,

$$\frac{Q_h}{T_h} = \frac{Q_c}{T_c},$$
$$\text{COP}_{\text{C}} = \frac{Q_c}{W_{\text{in}}} = \frac{T_c}{T_h - T_c}.$$

The coefficient of performance （性能系数） is not the same as a heat engine's thermal efficiency and can exceed one.

The second law forbids heat moving from cold to hot as the *sole* net effect. A refrigerator has another effect: consuming work. Also, work input does not have to generate entropy by itself; the ideal reversible refrigerator has zero total entropy production. Real devices generate positive entropy through irreversibilities.

Remember. Cold-to-hot heat pumping is permitted when driven by work. Check the complete cyclic energy and entropy balances.

QUESTION 22 / 26

Orientational disorder and residual entropy in solid CO

Does solid carbon monoxide (CO) have zero entropy at absolute zero (0 K) according to Third-Law measurements?

Answer: No

The intended answer concerns orientationally disordered solid carbon monoxide. A CO molecule can occupy similar lattice positions in two opposite orientations. Cooling can freeze in this orientational disorder （取向无序） before a perfectly ordered state is reached.

An idealized model with two independent equally likely orientations at each of N sites gives

$$\begin{aligned} W_0 &= 2^N, \\ S_{\mathrm{res}} &= k_{\mathrm{B}} \ln(2^N) = N k_{\mathrm{B}} \ln 2, \\ S_{\mathrm{res,m}} &= R \ln 2 \approx 5.76\,\mathrm{J\,K^{-1}\,mol^{-1}}. \end{aligned}$$

This count illustrates the mechanism; correlations and partial ordering can alter a real sample's value. Residual entropy is the entropy associated with the frozen configurational multiplicity, rather than a nonzero heat capacity at exactly zero kelvin. [R1]

The third-law zero-entropy statement requires the relevant perfectly ordered reference crystal and a unique ground state. It should not be applied indiscriminately to a kinetically trapped disordered sample. A calorimetric integral initialized at zero can miss a residual-entropy offset.

Remember. Do not equate "solid" with "a unique microstate." Orientational disorder can persist after translational motion has frozen out.

QUESTION 23 / 26

Raising the thermodynamic efficiency limit

Is the thermodynamic efficiency of a heat engine enhanced by operating with a hotter source and a colder sink?

Answer: Yes

For a reversible engine between two reservoirs,

$$\eta_{\mathrm{C}} = 1 - \frac{T_c}{T_h}.$$

Its dependence on the two absolute temperatures is explicit:

$$\left(\frac{\partial \eta_{\mathrm{C}}}{\partial T_h}\right)_{T_c} = \frac{T_c}{T_h^2} > 0,$$
$$\left(\frac{\partial \eta_{\mathrm{C}}}{\partial T_c}\right)_{T_h} = -\frac{1}{T_h} < 0.$$

Thus raising the hot-source temperature or lowering the cold-sink temperature increases the maximum possible efficiency.

The relevant quantity is the temperature *ratio*, not the temperature difference by itself. Two pairs with the same difference can have different limits:

$$(T_h, T_c) = (600, 300)\,\mathrm{K}: \qquad \eta_{\mathrm{C}} = 0.50,$$
$$(T_h, T_c) = (900, 600)\,\mathrm{K}: \qquad \eta_{\mathrm{C}} \approx 0.333.$$

Both differences are 300 K. A real engine satisfies $\eta \leq \eta_{\mathrm{C}}$; temperature changes raise the bound but do not guarantee that a redesigned real machine reaches it.

Remember. A hotter source and a colder sink improve the Carnot limit. Avoid replacing the ratio argument with a temperature-difference-only rule.

QUESTION 24 / 26

Integrating the Debye law gives one third of the heat capacity

Is the Third-Law entropy at low temperatures proportional to $C_{p,m}$ via $S_m(T) = \frac{1}{3}C_{p,m}(T)$ when the Debye T^3-law applies?

Answer: Yes

Assume the low-temperature Debye form applies from zero up to the temperature of interest, with no phase transition and zero residual entropy:

$$C_{p,\mathrm{m}}(T') = aT'^3, \qquad S_{\mathrm{m}}(0) = 0.$$

The third-law entropy follows by integration:

$$\begin{aligned} S_{\mathrm{m}}(T) &= \int_0^T \frac{C_{p,\mathrm{m}}(T')}{T'}\,\mathrm{d}T' \\ &= a\int_0^T T'^2\,\mathrm{d}T' \\ &= \frac{aT^3}{3} \\ &= \boxed{\frac{1}{3}C_{p,\mathrm{m}}(T)}. \end{aligned}$$

Both sides have the same molar entropy units, $\mathrm{J\,mol^{-1}\,K^{-1}}$. The result is a consequence of the power law, not a universal identity between entropy and heat capacity. [R2]

If a residual entropy is present, the corrected expression is

$$S_{\mathrm{m}}(T) = S_{\mathrm{res,m}} + \frac{1}{3}C_{p,\mathrm{m}}(T).$$

More generally, $C = aT^r$ with $r > 0$ gives $S(T) - S(0) = C(T)/r$ in its range of validity.

Remember. The factor $1/3$ comes from integrating T'^2. It requires the cubic heat-capacity law and the specified entropy reference.

QUESTION 25 / 26

Expansion makes translational quantum levels denser

Does expanding a gas into a larger volume cause its quantum energy levels to come closer together, increasing the number of accessible microstates?

Answer: Yes

For a particle of mass m in a three-dimensional cubic box of side length L, the translational energy levels （平动能级） are

$$E_{n_x n_y n_z} = \frac{h^2}{8mL^2}(n_x^2 + n_y^2 + n_z^2), \qquad n_x, n_y, n_z = 1, 2, \ldots$$

For fixed quantum labels, both the energies and characteristic spacings scale as

$$E \propto L^{-2} \propto V^{-2/3}.$$

Increasing volume therefore makes more translational states thermally accessible at a fixed temperature. In the classical dilute-gas limit, the one-particle translational partition function （平动配分函数） is

$$q_{\text{trans}} = \frac{V}{\Lambda^3}, \qquad \Lambda = \frac{h}{\sqrt{2\pi m k_{\text{B}} T}}.$$

At fixed T, q_{trans} is proportional to volume. For N noninteracting particles, the volume dependence of the many-particle count is proportional to V^N, yielding

$$\Delta S = N k_{\text{B}} \ln\left(\frac{V_f}{V_i}\right) = nR \ln\left(\frac{V_f}{V_i}\right).$$

Remember. This is the microscopic explanation of Question 4. It concerns translational confinement; expansion does not necessarily reduce every internal electronic or vibrational energy spacing.

QUESTION 26 / 26

When Gibbs and Helmholtz changes are equal

Is the Gibbs energy change ΔG equal to the Helmholtz energy change ΔA for any physical process involving ideal gases?

Answer: No

Start from the definitions, which hold independently of ideal-gas behavior:

$$G = H - TS, \qquad H = U + pV,$$
$$A = U - TS,$$
$$G = A + pV.$$

Therefore,

$$\boxed{\Delta G - \Delta A = \Delta(pV).}$$

For an ideal gas with fixed amount n,

$$pV = nRT \quad \Longrightarrow \quad \Delta G - \Delta A = nR(T_f - T_i).$$

A temperature change is an immediate counterexample to the claim "for any physical process." For example, heating 1.00 mol of ideal gas from 300 to 400 K gives a difference of approximately 831 J.

Correction to the original feedback. Changes in pressure or volume alone do *not* necessarily make $\Delta(pV)$ nonzero. For a fixed amount of ideal gas undergoing an isothermal expansion or compression,

$$\Delta(pV) = nR\Delta T = 0 \quad \Longrightarrow \quad \Delta G = \Delta A.$$

Remember. The general equality condition is $\Delta(pV) = 0$. For fixed-n ideal gases this is $\Delta T = 0$; if n changes, use $\Delta(nRT)$ instead.

A compact map of the main results

$$
\begin{aligned}
\Delta S_{\text{isolated}} &= S_{\text{gen}} \geq 0, \\
\Delta S_{\text{ideal gas},T} &= nR \ln\left(\frac{V_f}{V_i}\right), \\
\Delta S_V &= \int_{T_i}^{T_f} \frac{C_V(T)}{T}\, \mathrm{d}T, \\
\Delta S_{\text{trs,eq}} &= \frac{\Delta H_{\text{trs}}}{T_{\text{trs}}}, \\
\eta &\leq 1 - \frac{T_c}{T_h}, \\
\Delta A &= -T\Delta S_{\text{tot}} \quad (T, V \text{ fixed}), \\
\Delta G &= -T\Delta S_{\text{tot}} \quad (T, p \text{ fixed}), \\
\Delta G - \Delta A &= \Delta(pV).
\end{aligned}
$$

The two free-energy/total-entropy identities assume the closed-system reservoir setting and pressure-volume work only, as explained in Questions 8 and 17.

Source and supplementary reading

Quiz source. *3 Quiz yes/no entropy*, in *Thermodynamics, Ch. 1,2,3*, visible Notebook quiz and answer feedback, read on 28 October 2026.
https://notebook.google.com/notebook/982933e4-76f6-431d-b28f-bc67d4369556

[**R1**] E. Kozliak and F. L. Lambert, "Residual Entropy, the Third Law and Latent Heat," *Entropy* **10** (2008), 274–284. Used for the distinction between frozen configurational disorder and the ordered equilibrium reference; see Questions 7 and 22.
https://doi.org/10.3390/e10030274

[**R2**] David Tong, *Lectures on Statistical Physics*, Section 3.3.3, "The Debye Model." University of Cambridge. Used for the low-temperature $C_V \propto T^3$ result and its range of validity; see Questions 12 and 24.
https://www.damtp.cam.ac.uk/user/tong/statphys/statmechhtml/S3.html

Editorial clarification. The original quiz's Question 26 feedback suggests that a change in pressure or volume necessarily changes pV. That inference is false for a fixed amount of ideal gas undergoing an isothermal process. Question 26 here gives the corrected criterion. The work/entropy and refrigerator explanations also distinguish entropy production from work input itself.